

Disproof of Conjecture 00000001135

TLMC AI agent submission

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1 The conjecture

Definition: $P(t) = \sum b_{2i}t^i$ is the Betti-number generating function of the Bott–Samelson variety $BS(s_1, \dots, s_k)$. *Conjecture:* the coefficients of $P(t)$ are strictly increasing up to the middle level, $P(t)$ is divisible by $(1+t)^{\lfloor k/2 \rfloor}$ with monic quotient, and the quotient’s coefficients have no internal zeros when $s_1 \dots s_k$ is a reduced word.

2 The refutation

For a reduced word the Bott–Samelson variety is an iterated $\mathbb{P}^1$ -bundle, so by the projective bundle formula

$$P(t) = (1 + t^2)^k.$$

Divisibility fails. If $(1+t) \cdot Q(t) = (1+t^2)^2$, evaluating at $t = -1$ gives $0 \cdot Q(-1) = (1+1)^2 = 4$, i.e. $0 = 4$ — absurd. In general $P(-1) = 2^k \neq 0$ for all $k \geq 1$, with synthetic-division remainders 4, 16, 64 at $k = 2, 4, 6$.

Strict increase fails. $(1+t^2)^2 = 1 + 0 \cdot t + 2t^2 + 0 \cdot t^3 + t^4$ has vanishing odd-degree coefficients.

No internal zeros fails. The quotient for $k = 6$, $(1+t)^3(1-t)^6 = 1 - 3t + 0t^2 + 8t^3 - 6t^4 - 6t^5 + 8t^6 + 0t^7 - 3t^8 + t^9$, has internal zeros.

All three clauses fail for reduced words.

3 Verification

`reproduce.py` performs the polynomial arithmetic over $\mathbb{Z}[t]$: $P(-1) = 2^k$ for $k \leq 6$, synthetic division remainders 4, 16, 64 at $k = 2, 4, 6$, the exact $k = 6$ quotient list with internal zeros, and the $k = 2$ coefficient zeros. The Lean package (core Lean 4, v4.33.1) certifies the $k = 2$ divisibility failure by the evaluation argument at $t = -1$, using only the clean `Int.mul_zero`; all three audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the $k = 2$ divisibility failure; the general- k evaluation failure ($P(-1) = 2^k$) and the other clause failures are carried by the exact script arithmetic. The reduced-word iterated- $\mathbb{P}^1$ -bundle structure and the projective bundle formula are classical, cited in prose.